minutes. At first he suspects a mechanical problem, since those times seem to indicate that the car was traveling slower than 100 miles per hour on the last three runs. Yet after examining the car extensively he convinces himself that it is in perfect condition. Unable to explain the anomalously long times, he consults Slim and asks him about the final few runs. Slim has a simple explanation. He tells Jim that, since the track runs from east to west, as it got later in the day, the sun was glaring into his view. During the last three runs it was so bad that he drove from one end of the track to the other at a slight angle. He draws a rough sketch of the path he took on the last three runs, and it is shown in Figure 2.5. The explanation for the three longer times is now perfectly clear: the path from start to finish is longer when traveling at an angle and therefore, at the same speed of 100 miles per hour, it will take more time to cover. Put another way, when traveling at an angle, part of the 100 miles per hour is expended on going from south to north, leaving a bit less to accomplish the trip from east to west. This implies that it will take a little longer to traverse the strip.

As stated, Slim's explanation is easy to understand; however, it is worth rephrasing it slightly for the conceptual leap we are about to take. The north-south and east-west directions are two independent spatial dimensions in which a car can move. (It can also move vertically, when traversing a mountain pass, for example, but we will not need that ability here.) Slim's explanation illustrates that even though the car was traveling at 100 miles per hour on each and every run, during the last few runs it shared this speed between the two dimensions and hence appeared to be going slower than 100 miles per hour in the east-west direction. During the previous runs, all 100 miles per hour were devoted to purely east-west motion; during the last three, part of this speed was used for north-south motion as well.

Einstein found that precisely this idea—the sharing of motion between different dimensions—underlies all of the remarkable physics of special relativity, so long as we realize that not only can spatial dimensions share an object's motion, but the time dimension can share this motion as well. In fact, in the majority of circumstances, most of an object's motion is through time, not space. Let's see what this means.